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Disassembly sequence planning for target parts of end-of-life smartphones using Q-learning algorithm

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Abstract

Owing to increasing environmental concerns, disassembly has become an important step in value recovery from end-of-life (EoL) products. As one of the prevalent electronic devices, smartphones are especially challenging to manage at the end of life. They have a diverse internal structure that leads to difficulties in modeling and developing general plans for disassembly. Given this background, an improved method that uses a Q-learning algorithm is proposed to optimize the disassembly sequence of EoL smartphones. A constraint relationship is first developed of EoL smartphone parts. The disassembly sequence planning problem is then formalized with a Markov Decision Process. The optimization objective is the disassembly time to obtain the target parts. The disassembly time and the target parts are converted into a reward value for the Q-learning algorithm. A State-Action-Reward Matrix is established based on converted reward value and disassembly sequence planning problem. A Q-table is trained to find the best action for each state using the State-Action-Reward Matrix. The disassembly sequence with the maximum reward value is secured through the trained Q-table and objective function. A case study of the 'Xiaomi 5' is conducted to verify the applicability of the proposed method. The experimental results demonstrate that the proposed method can provide feasible sequence planning for targeted parts in the disassembly of EoL smartphones.

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1. Introduction

The idea of recycling and reusing end-of-life (EoL) products is increasingly important in terms of circular economy [1]. Recovering value from EoL products is important from an environmental perspective as well as from an economic and resource saving viewpoint [2]. Disassembly is regarded as a key link in the recycling process [3]. A reasonable disassembly sequence can not only reduce disassembly time and increase profit, but also protect the environment [4]. Therefore, it is essential to use an efficient optimization method to obtain a satisfactory disassembly sequence.

At present, heuristic optimization algorithms are used to obtain disassembly sequences by a majority of researchers. Such approaches often struggle in practice because EoL smartphones have a diverse internal structure that leads to difficulties in modelling and developing general purpose plans for disassembly. Therefore, a new method is needed, e.g. utilizing machine learning that can find suitable disassembly sequences/plans.

Given this background, the Q-learning algorithm is proposed. It offers simplicity and wide applicability. One challenge in applying the Q-learning algorithm to the disassembly sequence planning problem is obtaining a realistic reward value. In this paper, a disassembly sequence planning method for target parts of EoL smartphones using Q-

learning algorithm is proposed. The reward value of the Q-learning algorithm is obtained based on disassembly time and the target parts. In comparison with other studies, this paper has the following contributions: 1) The Q-learning algorithm is applied to the disassembly sequence planning problem for target parts of EoL smartphones. This represents a new application for the Q-learning algorithm. The disassembly sequence planning process for target parts of EoL smartphones is simplified. 2) The new reward function of the Q-learning algorithm is built based on the optimization objective (the disassembly time and the target part) and the results match well with the experiments.

2. Literature review

The disassembly sequence planning problem has attracted the interest of many researchers and different algorithms have been used to solve the disassembly sequence planning problem [5]. Yeh et al. [6] proposed a simplified swarm optimization method for disassembly sequence planning. Kongar et al. [7] used genetic algorithms to solve the disassembly sequence planning problem of EoL products. In order to solve the problems within discrete solution spaces and with complex disassembly precedence constraints, Xia et al. [8] presented a simplified teaching-learning-based optimization algorithm. Tseng et al. [9–11] proposed a flatworm algorithm, a block-based genetic algorithm and a hybrid bidirectional ant colony optimization algorithm to optimize the disassembly sequence. As the size of components increases, exact algorithms struggle to obtain the optimal disassembly sequence. In order to solve the problem, Xie et al. [12] proposed a promising method using modified grey wolf optimizer. Xu et al. [13] used a modified discrete bees algorithm based on a Pareto front for disassembly sequence planning in remanufacturing. Reinforcement learning algorithms are also growing in use. Sugisawa et al. [14] proposed machining sequence learning via inverse reinforcement learning. Zhang et al. [15] proposed autonomous generation of action sequence based on reinforcement learning. Qu et al. [16] proposed novel reinforcement learning based grey wolf optimizer algorithm for unmanned aerial vehicles path planning. Maoudj et al. [17] proposed an optimal path planning approach based on Q-learning algorithm for mobile robots.

Considering the different characteristics of different products, in-depth research on the sequence planning of specific EoL products has been conducted. Go et al. [18] used a genetic algorithm to optimize disassembly sequence for automotive component reuse. Kheder et al. [19] proposed an optimized disassembly process based on a genetic algorithm to find the best feasible disassembly sequence for automotive rear axles. Jin et al. [20] used particle swarm optimization to search the optimized selective disassembly sequence for liquid crystal display televisions. Different methods are used to assess the disassembly time. Aicha et al. [21] build the disassembly time model to calculate the disassembly time. Favi et al. [22] uses big data analysis for the estimation of

disassembly time and de-manufacturing activity. Marconi et al. [23] presents a method for calculating the effective disassembly sequence and time for industrial products by data mining technique.

The above research results have laid the foundation for research on disassembly sequence planning. As one of the prevalent electronic devices, end-of-life smartphones are difficult to model and general purpose disassembly plans are hard to develop. A Q-learning algorithm can avoid the above difficulties in the disassembly sequence optimization. Therefore, a Q-learning algorithm with a realistic reward function is employed for disassembly sequence planning and then verified by disassembling EoL smartphones. The new reward function of the Q-learning algorithm is proposed based on the disassembly time and the target part.

3. Disassembly sequence planning method

In this section, an improved disassembly sequence planning method with a Q-learning algorithm is introduced. In the first step, the disassembly environment is built based on the constraint relationship of EoL smartphone parts. The second step is to formalize the disassembly sequence planning problem for target parts of EoL smartphones with a Markov Decision Process (MDP). In the third step, the disassembly time model of EoL smartphones is established. In the fourth step, the disassembly time and the state whether the target part is disassembled are converted into a realistic reward value for the Q-learning algorithm. The fifth step is to establish a State-Action-Reward Matrix. In the sixth step, a Q-table is trained to find the best action in each state by using the State-Action-Reward Matrix. The disassembly sequence with the maximum reward value is identified through the trained Q-table and objective function. The flow chart of the method using Q-learning algorithm is shown in Fig.1.

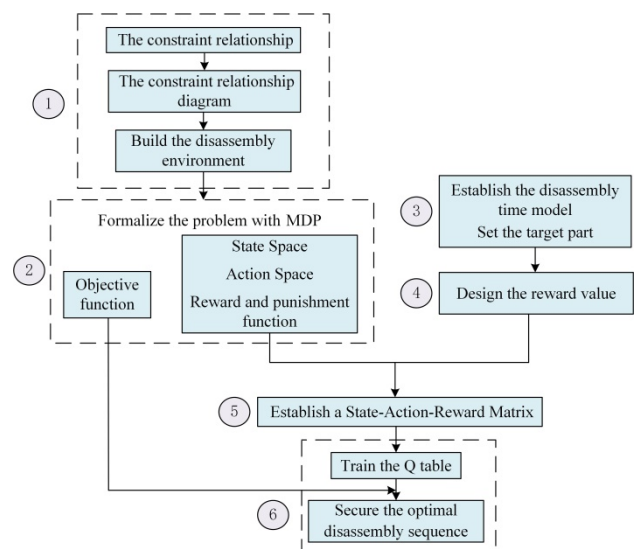


Fig. 1. The flow chart of disassembly sequence planning method using Q-learning algorithm.

The disassembly profit is increased by reducing the disassembly time. Accurate assessment of disassembly time

has a significant influence on the disassembly sequence. Moreover, the reward is a very important element of a Q-learning algorithm. The more realistic the reward design is, the better the algorithm will be at producing results that match actual outcomes. In disassembly sequence planning for target parts of EoL smartphones, the performance or fitness is the disassembly time to obtain the target part. So, for the proposed method, the reward is designed based on the disassembly time and the state whether the target part is disassembled. The disassembly time model is established to calculate the disassembly time of each part. The shorter the disassembly time is, the higher the reward value is. In order to avoid the disassembling parts with constraints, the punishment value should be higher than the reward value of disassembly time. Moreover, the reward value is obtained by the state whether the target part is disassembled. Finally, the reward of each part is obtained based on the reward values of the state whether the target part is disassembled and the disassembly time.

3.1. The disassembly environment

For product disassembly, constraint relationship diagram is commonly used. The component is one part or a combination of multiple parts that can be directly recycled [1]. In this paper, the component is assumed as a part that does not require further disassembly. Precedence constraint between two parts is connected by unidirectional edge. For example, Fig. 2 shows the constraint relationship diagram between the parts of EoL smartphones. Part A is the predecessor of Part B, which means that Part A is disassembled before Part B. Part E is the predecessor of Part D, which means that Part E is disassembled before Part D.

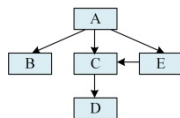


Fig. 2. The constraint relationship diagram between the parts of EoL smartphones.

The disassembly environment is built based on the constraint relationship diagram in this paper. First of all, the disassembly sequence planning problem for target parts of EoL smartphones is assumed as a breakthrough game. The parts of EoL smartphones are set as the game level, the target part is set as the target level of the game. If Part A is the predecessor of Part B, it means that the level A is passed before the level B. If Part A and Part E are the predecessor of Part C, it means that the level A and the level E are passed before the level C. The goal of the game is to get the target part in the shortest time. The reward value of subsequent levels will be attenuated when a previous level is passed.

3.2. Formalize disassembly sequence planning problem with MDP

The disassembly sequence planning problem for target parts of EoL smartphones is to find an optimal disassembly action in each disassembly state for obtaining the target parts. Therefore, the disassembly sequence planning problem for

target parts of EoL smartphones is a sequential decision problem. The Markov Decision Process is used to solve the sequential decision problem. The formalized representation of sequence planning problem for target parts of EoL smartphones is as follows.

State space S :

$$S = [S_0, S_1, S_2, S_3, S_4 \cdots S_n] \quad (1)$$

where S_n ($n=0,1,2,3\cdots$) is the disassembly state in which part n will be disassembled.

Action space D :

$$D = [D_0, D_1, D_2, D_3, D_4 \cdots D_n] \quad (2)$$

where D_n ($n=0,1,2,3\cdots$) is the disassembly action in which part n will be disassembled.

Reward and punishment function R :

$$R = [q, w, e] \quad (3)$$

where q is a punishment value when the disassembly part has constraints, w is a reward value when the disassembly part is not the target part and doesn't have constraints, e is a reward value when the disassembly part is the target part and doesn't have constraints.

Objective function: Q-learning algorithm defines discounted cumulative returns with the value function. The value function is defined as follows:

$$q_\pi(S, D) = E_\pi [G_n | S_n = S, D_n = D] \quad (4)$$

$$q_\pi(S, D) = E_\pi \left[\sum_{k=0}^{\infty} \gamma^k R_{n+k+1} | S_n = S, D_n = D \right] \quad (5)$$

where q is the state and action value function, E is the expected value in state S , S is the state, D is the action performed in S state, n is the number of the state, γ is the discount factor, k is the step size. π is the policy that the maximum reward value is obtained.

3.3. Establish disassembly time model

The disassembly time model is needed as the basis of the reward value. The disassembly time model includes tool changing time and auxiliary time for disassembly activities. The disassembly time model for the single part is as follows:

$$t_i = f_i + u_i + h_i \quad (6)$$

where t_i is the disassembly time of the part i , f_i is the basic disassembly time (the time of the necessary disassembly action) of the part i , u_i is the tool changing time of the part i , h_i is the auxiliary time of the part i , i is part number.

The tool changing time of the single part is as follows:

$$u_i = N_i u_i \quad (7)$$

where u_i is the time of changing the disassembly tool once, N_i is the number of changing tools when the part i is disassembled. If the disassembly tools are not changed, $N_i=0$. If the disassembly tools are changed once, $N_i=1$. If the disassembly tools are changed n times, $N_i=n$.

The auxiliary time of the single part is as follows:

$$h_i = Z_i M_i + C_i \quad (8)$$

where M_i is the positioning time of the tools when the constraints are removed from the part, Z_i is the number of

constraints of part i , C_i is the time of tidying up each part.

Based on the disassembly time model of the single part, the total disassembly time of EoL smartphones is as follows:

$$T = \sum_{i=1}^n t_i = \sum_{i=1}^n f_i + \sum_{i=1}^n N_i u_i + \sum_{i=1}^n (Z_i M_i + C_i) \quad (9)$$

where T is the total disassembly time.

3.4. Design the reward value

In order to obtain good result, the reward value is designed based on the disassembly time and the state whether the target part is disassembled. The total reward value is calculated by equation (10). If a part does not have constraints and can be disassembled, the total reward value R of the single part is as follows:

$$R = R(t) + R(m) \quad (10)$$

where $R(m)$ is the reward value of the state whether the target part is disassembled. $R(t)$ is the reward value based on disassembly time, which is calculated by equation (11).

$$R(t) = \frac{Z}{t_i} \quad (11)$$

where t_i is the disassembly time of each part, Z is the correction factor and is set to 100.

To sum up, the total reward value of the single part is as follows:

$$R = \frac{Z}{t_i} + R(m) \quad (12)$$

3.5. Establish State-Action-Reward Matrix

The states, actions and reward value are expressed as a State-Action-Reward Matrix. The rows of the matrix are states of each part and the columns of the matrix are possible disassembly actions in each state. The elements of the matrix are the reward or punishment value of executing a possible action in a state. Based on the constraint relationship between the parts in Fig.2, the matrix is built as shown in Fig.3. P is the punishment value in Fig.3.

$$H = \begin{pmatrix} P & R(t)_B & P & P & R(t)_E \\ P & P & P & P & R(t)_E \\ P & R(t)_B & P & R(t)_D + R(m) & P \\ P & R(t)_B & P & P & P \\ P & R(t)_B & R(t)_C & P & P \end{pmatrix}$$

Fig. 3. The State-Action-Reward Matrix H for Fig. 2

For example, third row and third column of the matrix is P . It means that the part C has constraints and can't be disassembled in the state that the part B has been disassembled. Third row and second column of the matrix is $R(t)_B$. It means that the part B does not have constraints and can be disassembled in the state that the part C has been disassembled. Third row and fourth column of the matrix is $R(t)_D + R(m)$. It means that the part D without constraints can be disassembled and the part D is target part in the state that part C has been disassembled.

3.6. Train Q-table and obtain optimal disassembly sequence

Q-table is trained based on a State-Action-Reward Matrix, and is constantly updated during training. The Q value update formula is as follows:

$$Q(S_t, D_t) \leftarrow (1-\alpha) Q(S, D) + \alpha [r_{t+1} + \gamma \max_{a_{t+1}} Q(S_{t+1}, D_{t+1})] \quad (13)$$

where S_t is the current state, D_t is the action performed in S_t state, r_{t+1} is the reinforcement signal after S_t executed, S_{t+1} is the next state, D_{t+1} is the action performed in S_{t+1} state, γ is the discount factor, α is the learning coefficient.

The Q-table that finds the best action for each state is obtained through training. The optimal disassembly sequence with the maximum reward value is secured based on the objective function in Section 3.2 and trained Q table. The sequence generation process is shown in Fig.4.

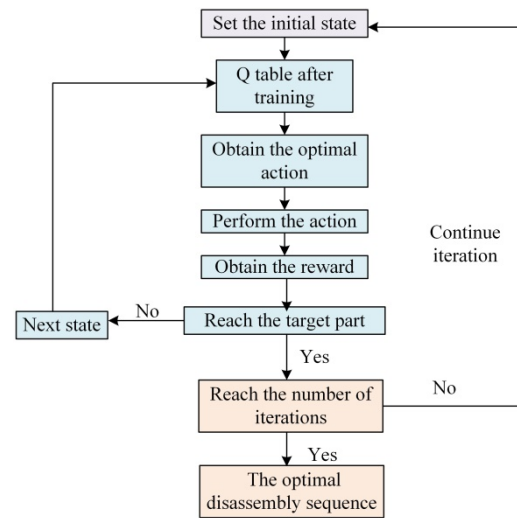


Fig. 4. The sequence generation process.

4. Case study

The applicability of the proposed method is verified through the case study. It is assumed that the disassembly time of the parts is fixed. EoL smartphones, namely Xiaomi 5, was selected as an example. The disassembly parts diagram of Xiaomi 5 is shown in Fig.5. The constraint relationship diagram of Xiaomi 5 is shown in Fig.6. The basic disassembly time is obtained by disassembly experiments. The basic disassembly information parameters of Xiaomi 5 are listed in Table 1.

The tool changing time is 4 seconds when the disassembly tools are changed once. The positioning time of the tools is 1 second when removing a constraint from the part. The time of tidying up a part is 5 seconds. If the target part is disassembled, $R(m)$ is set to 200. If the target part is not disassembled, $R(m)$ is set to 0. Moreover, the punishment value is set to -50 for avoiding the disassembling parts with constraints. The motherboards of EoL smartphones contain a lot of non-renewable metals and have high recycling value. Therefore, the motherboard is selected as the target part. The total disassembly time and the total reward value of each part are shown in Table 2.

Table 1. Basic disassembly information parameters of Xiaomi 5

Part No.	Part name	Disassembly tool	Constraints/Quantity	Basic time (s)
1	Phone cover	Sucker, crowbar	Buckle/1	12
2	Motherboard cover	Screwdriver, tweezers	Screws/9, buckle/1	26
3	Tail plate cover	Screwdriver, tweezers	Screws/7, buckle/1	20
4	RF line	Tweezers	Embedded/1	7
5	Battery	Hot air gun, tweezers, sucker	Sealant/1, buckle/1, limit/1	50
6	Back camera cover	Crowbar, tweezers	Limit/1	5
7	Back camera	Tweezers	Buckle/1	4
8	Motherboard	Screwdriver, tweezers	Screws/1, limit/1	15
9	Front camera	Tweezers	Buckle/1	5
10	Tail plate	Screwdriver, crowbar	Screws/2, buckle/1	12
11	Screen module	Hot air gun, crowbar, tweezers, sucker	Sealant/1, limit/1, buckle/1	148

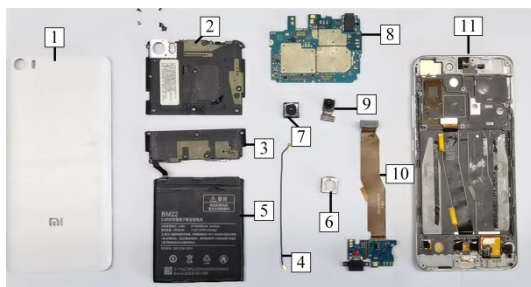


Fig. 5. The disassembled parts diagram of Xiaomi 5

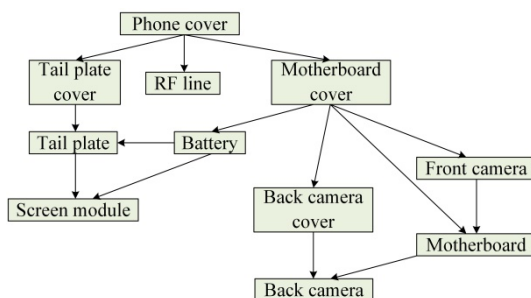


Fig. 6. The constraint relationship diagram of Xiaomi 5

Table 2. The total disassembly time and the reward value of each part

Part No.	Time (s)	Reward value
1	27	3.70
2	49	2.04
3	41	2.44
4	17	5.88
5	81	1.23
6	20	5.00
7	14	7.14
8	30	203.33
9	15	6.67
10	28	3.57
11	172	0.58

4.1. Results

In this paper, the proposed method is implemented in MATLAB 2019b and runs on a PC with an Intel i7-11800H Core CPU at 2.3 GHz and 16 GB of RAM. In the proposed method, the update step is 1, the discount factor is 0.8 and the number of iterations is 50. When the target part is obtained, the disassembly sequence with the maximum reward value is decided among all feasible sequences by using objective function in Section 3.2. The best disassembly sequence is 1-2-9-8 (phone cover - motherboard cover - front camera - motherboard) and the corresponding total disassembly time is 121 seconds in this case.

4.2. Validation of the proposed method

The disassembly experiment is conducted to verify the feasibility of the proposed method and the disassembly time model. It is assumed that the damage degree of EoL smartphones used in the experiment is same, and the disassembly proficiency level of the operators is similar. EoL smartphones are disassembled without damage in the disassembly experiment. Three operators with similar disassembly experience are selected, and the disassembly experiments of every operator are recorded by a video camera and repeated three times. The disassembly time of three operators with their own experience is recorded for the motherboard remove of Xiaomi 5 (Group A). Then, each operator of group A under the guidance of the proposed method repeated the disassembly experiment with same procedure and results were obtained (Group B). The average disassembly time of each operator and average disassembly time of three operators are calculated in each group. The error of the disassembly time is calculated to judge whether the reward of proposed method is realistic and the disassembly time model can calculate the disassembly time accurately. The disassembly experimental results for verifying the feasibility of the reward of the proposed method and the disassembly time model are shown in Table 3.

The average disassembly time of three operators in Group B decrease 17.9% compared with the result in Group A. Compared to average disassembly time of three operators in Group B, the error of the disassembly time model is 0.64%. The experimental results demonstrate that the proposed method can reduce disassembly time and the reward of the proposed method and the disassembly time model is realistic. Therefore, the proposed method can provide optimized sequence planning for target parts disassembly of EoL smartphones.

Although we have successfully applied the proposed method in smartphones, there are some limitations in this method. Firstly, it is important to consider the problem of the uncertain disassembly time. The disassembly times were obtained by manual disassembly experiment and the average disassembly time is calculated to reduce the uncertainty in the study. Then, EoL smartphones, namely Xiaomi 5, was selected as one example of multiple waste electronic products

in this paper. In order to determine the application range of the proposed method, it is meaningful to study other waste electronic products.

Table 3. The disassembly experiment results

Operators	Disassembly sequence	Average disassembly time of each operator(s)	Average disassembly time of three operators(s)
A	1	1-2-10-4-9-8	153.33
	2	1-2-6-9-8	138.66
	3	1-2-10-4-9-8	153.33
B	1	1-2-9-8	121.67
	2	1-2-9-8	124.00
	3	1-2-9-8	119.67

5. Conclusions

End-of-life smartphones have a diverse internal structure that leads to modeling difficulties and difficulties in developing general purpose plans for disassembly. In order to solve the problem, the improved disassembly sequence planning method for target parts of EoL smartphones using Q-learning algorithm is proposed. Firstly, the disassembly environment of EoL smartphones is built based on the constraint relationship diagram. In addition, the disassembly sequence planning problem is formalized with a Markov Decision Process based on the disassembly environment. In order to get the target part in the shortest time, the disassembly time and the state whether the target part is disassembled are converted into the reward value of Q-learning algorithm. Finally, a State-Action-Reward Matrix is established based on the converted reward value and the formalized disassembly sequence planning problem. A Q-table is trained to find the best action in each state by using the State-Action-Reward Matrix. The disassembly sequence with the maximum reward value is secured through the trained Q-table and objective function. A case study of the 'Xiaomi 5' is conducted to verify the feasibility of the proposed method. The experimental results demonstrate that the proposed method can provide optimized sequence planning for target parts disassembly of EoL smartphones.

This paper is a preliminary exploration on the application of Q-learning algorithm in disassembly sequence planning for target parts of EoL smartphones. The limitations will be the future research direction.

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